9.4 Independent and Dependent Events

Name _____ Period ____ Date _____

Algebra I 9.4 (30-minute plan) · Student sheet

I can…

- tell whether two events are independent or dependent.
- find the probability of independent events.
- find the probability of dependent events.

Vocabulary · fill in the missing word

Independent events — the _____ of one event does not affect the other

Dependent events — the result of one event changes the _____ of the other

Independent — P(A and B) = _____

Dependent — P(A and B) = _____

With replacement — the item is put back, so the events are _____

Without replacement — the item is kept, so the events are _____

Part 1 · Notes and examples (fill in each note, then work the example)

Independent: one event _____ the probability of the other. P(A and B) = _____

Replacement: With replacement → _____. Without replacement → _____.

Example 1

Tell whether the events are independent or dependent.

a. You roll a die and flip a coin.

b. You draw a card, don't put it back, and draw another card.

c. A class elects a president, then elects a vice president from the other students.

Example 2

You spin a spinner with four equal sections (red, blue, yellow, green) and roll a die. What is the probability of red and a 6?

Dependent: P(A and B) = P(A) · _____

Without replacement: Without replacement, the second fraction has one fewer in the _____ and in the _____.

Example 3

A drawer has 4 black socks and 6 white socks. You pick two socks at random. Find P(both black)

a. with replacement. b. without replacement.

Part 2 · Partner practice

Partner practice

A jar has 5 red and 3 green candies. You eat one at random, then take another. What is the probability that both are green?